

EmbedUR Linux Training

Module - 2 : Shell Scripting

Assessment - 3

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Scenario: Automating file backup and Reporting to the system. Create a shell script called "backup_manager.sh" that performs the following tasks incorporating the concepts suggested.

Requirements:

1. Command-line Arguments and Quoting:

The script must accept three arguments: Source directory: A directory containing files to back up. Backup directory: The destination where files will be backed up. File extension: A specific file extension to filter (e.g., .txt).

Example: `./backup_manager.sh "/home/user/source" "/backup" ".txt"`

2. Globbing:

The script should use globbing to find all files in the source directory matching the provided file extension.

3. Export Statements:

Use export to set an environment variable `BACKUP_COUNT`, which tracks the total number of files backed up during the script execution.

4. Array Operations:

Store the list of files to be backed up in an array.

Print the names of these files along with their sizes before performing the backup.

5. Conditional Execution:

If the backup directory does not exist, create it. If creation fails, exit with an error.

If the source directory is empty or contains no files matching the extension, exit with a message.

If a file already exists in the backup directory with the same name, only overwrite it if it is older than the source file (compare timestamps).

6. Output Report:

After the backup, generate a summary report displaying:

Total files processed.

Total size of files backed up.

The path to the backup directory.

The report should be saved in the backup directory as `backup_report.log`.

Code:

```
#!/bin/bash
```

```
read source
```

```
read backup
```

```
read file_extn
```

```
cnt=$(find "$source" -maxdepth 1 -type f -name "$file_extn" | wc -l)
```

```
arr=("$source"/"$file_extn")
```

```
shopt -u nullglob
```

```
echo "Count: $cnt"
```

```
export cnt
```

```
echo "Files in source:"
```

```
for file in "${arr[@]}; do
```

```
    echo "$file"
```

```
done
```

```
if [ -d "$backup" ]; then
```

```
    echo "Backup path exists"
```

```
    arr_new=("$backup"/*."$file_extn")
```

```
    echo "Files in backup before copy:"
```

```
    for file in "${arr_new[@]}; do
```

```
        echo "$file"
```

```
    done
```

```
    echo "Copying files..."
```

```
    for file in "${arr[@]}; do
```

```
        cp "$file" "$backup"
```

```
    done
```

```
    cnt_backup=$(find "$backup" -maxdepth 1 -type f -name "*.$file_extn" | wc -l)
```

```
    echo "Backed Up Files Count: $cnt_backup"
```

```
    total_size=$(du -sh "$backup")
```

```
    echo "Total size of the backup dir: $total_size"
```

else

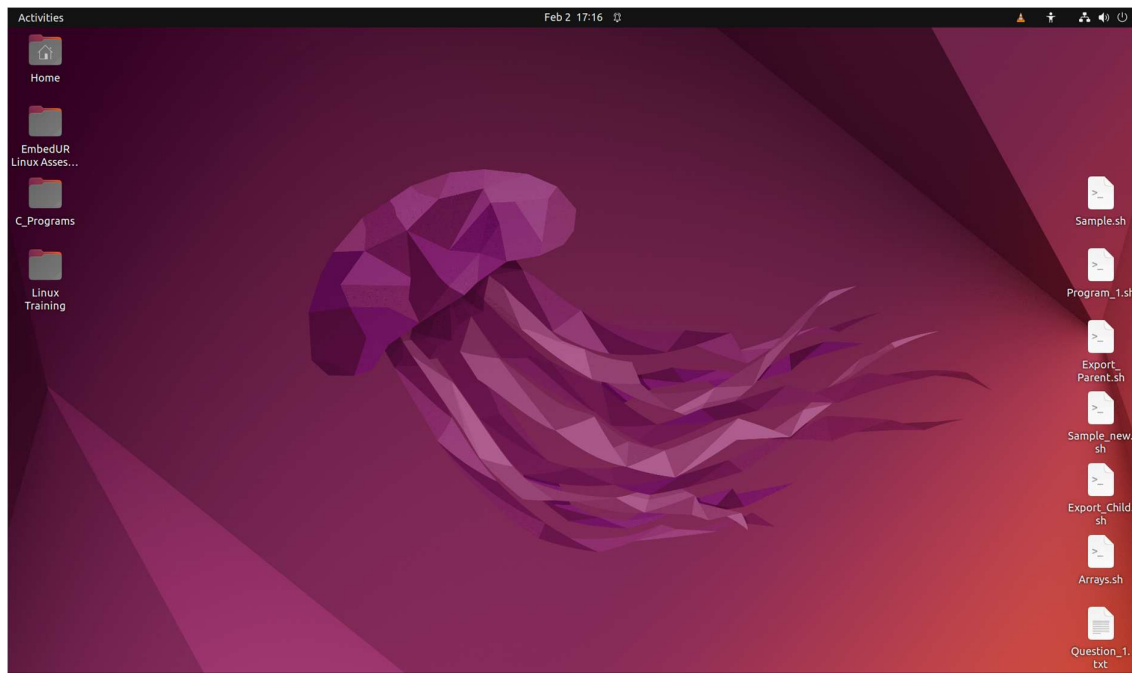
echo "Backup path does not exist"

fi

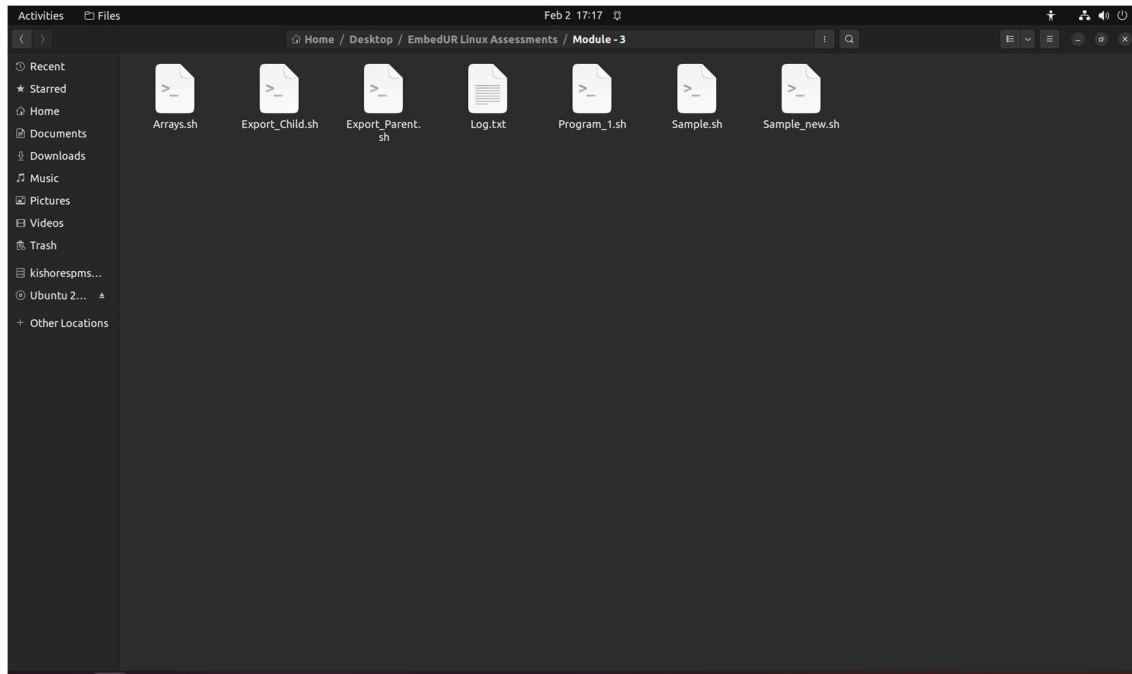
Outputs:

```
Activities Terminal Feb 2 17:14
kishore@kishore-virtual-machine: ~/Desktop/EmbedUR Linux Assessments/Module - 3
kishore@kishore-virtual-machine:~/Desktop/EmbedUR Linux Assessments/Module - 3$ bash Program_1.sh > Log.txt
/home/kishore/Desktop/EmbedUR Linux Assessments/Module - 3
/home/kishore/Desktop
sh
kishore@kishore-virtual-machine:~/Desktop/EmbedUR Linux Assessments/Module - 3$
kishore@kishore-virtual-machine:~/Desktop/EmbedUR Linux Assessments/Module - 3$ cat Log.txt
Count: 6
Files in source:
/home/kishore/Desktop/EmbedUR Linux Assessments/Module - 3/Arrays.sh
/home/kishore/Desktop/EmbedUR Linux Assessments/Module - 3/Export_Child.sh
/home/kishore/Desktop/EmbedUR Linux Assessments/Module - 3/Export_Parent.sh
/home/kishore/Desktop/EmbedUR Linux Assessments/Module - 3/Program_1.sh
/home/kishore/Desktop/EmbedUR Linux Assessments/Module - 3/Sample_new.sh
/home/kishore/Desktop/EmbedUR Linux Assessments/Module - 3/Sample.sh
Backup path exists
Files in backup before copy:
/home/kishore/Desktop/Arrays.sh
/home/kishore/Desktop/Export_Child.sh
/home/kishore/Desktop/Export_Parent.sh
/home/kishore/Desktop/Program_1.sh
/home/kishore/Desktop/Sample_new.sh
/home/kishore/Desktop/Sample.sh
Copying files...
Backed Up Files Count: 6
Total size of the backup dir: 1.3G /home/kishore/Desktop
kishore@kishore-virtual-machine:~/Desktop/EmbedUR Linux Assessments/Module - 3$
```

Files Backed Up in the Desktop



Source Directory



Backup Count: 6

Files in source:

/home/kishore/Desktop/EmbedUR Linux Assessments/Module - 3/Arrays.sh

/home/kishore/Desktop/EmbedUR Linux Assessments/Module - 3/Export_Child.sh

/home/kishore/Desktop/EmbedUR Linux Assessments/Module - 3/Export_Parent.sh

/home/kishore/Desktop/EmbedUR Linux Assessments/Module - 3/Program_1.sh

/home/kishore/Desktop/EmbedUR Linux Assessments/Module - 3/Sample_new.sh

/home/kishore/Desktop/EmbedUR Linux Assessments/Module - 3/Sample.sh

Backup path exists

Files in backup before copy:

/home/kishore/Desktop/Arrays.sh

/home/kishore/Desktop/Export_Child.sh

/home/kishore/Desktop/Export_Parent.sh

/home/kishore/Desktop/Program_1.sh

/home/kishore/Desktop/Sample_new.sh

/home/kishore/Desktop/Sample.sh

Copying files...

Backed Up Files Count: 6

Total size of the backup dir: 1.3G /home/kishore/DesktopLog: